Choose a model for the quantities and configurations your simulation must predict. Speed, accuracy, and transferability depend on the task; they are not universal scores.

1 What is evaluated?

Classical force field

positions → fitted energy formula → energy + forces

ML potential

local environments → trained neural model → energy + forces

Density functional theory

nuclei + electrons → self-consistent electrons → energy + forces

All three can supply energies and forces for atomistic simulation. They differ in how that mapping is built and what electronic physics is represented.

2 What must be validated?

Method	Checks for the intended application
Classical force field	Functional form and parameter coverage; bonding changes, charges, and long-range interactions.
ML potential	Training-domain coverage; errors on representative structures, forces, defects, and trajectories.
DFT	Functional, basis and sampling convergence; relevant charge, spin, and electronic states.

Benchmark the same observable, structures, and accuracy target on the intended hardware. Report throughput and errors with units; test transfer on configurations excluded from fitting.

Ask “good enough for this task?” before “which is best?” Classical and ML models amortize a fitted approximation over many evaluations. DFT solves an approximate electronic problem for each configuration. None guarantees accuracy outside its validated setting.